

Iterative Development of a Software to Facilitate Independent Home Use of BCI Technologies for Children with Quadriplegic Cerebral Palsy

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Abstract—Brain-computer interfaces (BCIs) are emerging as a new solution for children with severe disabilities to interact with the world. However, BCI technologies have yet to reach end-users in their daily lives due to significant translational gaps. To address these gaps, we applied user-centered design principles to establish a home BCI program for children with quadriplegic cerebral palsy. This work describes the technical development of the software we designed to facilitate BCI use at home. Children and their families were involved at each design stage to evaluate and provide feedback. Since deployment, seven families have successfully used the system independently at home and continue to use BCI at home to further enable participation and independence for their children.

Clinical relevance—The design and successful implementation of user-centered software for home use will both inform on the feasibility of BCI as a long-term access solution for children with neurological disabilities as well as decrease barriers of accessibility and availability of BCI technologies for end-users.

I. INTRODUCTION

Brain-computer interfaces (BCIs) are an emerging access technology, poised to help individuals with severe disability by enhancing, augmenting, or replacing volitional physical movement. BCIs record and interpret patterns in brain activity to generate control commands for operating a variety of applications [1]. Despite encouraging advancements in BCI development over the last several decades, few BCI technologies have reached end-users in their daily lives [2]. BCI is currently suffering from a critical translational gap between lab and home, due in part to the cost, reliability, and perceived intricacy of these systems for independent use in highly unique, complex end-user cases [2], [3]. It is crucial that BCI developers and researchers focus on how to make these technologies effective, available, and accessible for relevant environments for end-users and their caregivers.

The user-centered design (UCD) framework, which incorporates end-user needs right from the start of the design process, can be used to help bridge the BCI translational gap and maximize the success of BCI technologies for long-term use [4]. In the UCD framework, steps are taken to understand user needs and context of use, iterating through designs with evaluation from the users themselves at each cycle [4]. Several groups report on applying the UCD framework to BCI equipment and software design for home use by end-users with disabilities, including the *BackHome* system [5],

[6], the *Wadsworth BCI* system [7] and the *teleBCI* system [8]. All groups incorporated end-users in the planning and evaluation of their designs, reported early success, and will continue to refine these systems for their end-users.

Despite these initial advances for adult BCI end-users, the translational gap remains wide for pediatric end-users, such as children living with quadriplegic cerebral palsy (QCP). Little research has been done, even in the laboratory, to understand the feasibility of BCI for this population. We have previously demonstrated that both typically developing children and children with perinatal stroke can use simple BCI systems [9], [10] and have established a clinical BCI program providing disabled children access to BCI technologies [11] for applications like gaming [12] and exploring movement [13]. With the advent of the COVID-19 pandemic, we were forced to face the translational gap head-on and find a way to bring BCI technologies to children in their homes when the rest of the world had gone “online”. The rest of this paper describes the technical development and iterations of our home BCI software, directly informed at each stage from feedback from the children and families. We demonstrate how, through user-centered design, access to BCI software can provide greater opportunities for independence and participation for disabled children.

II. METHODS

A. Home BCI Pilot Program Overview

Families for the pilot home program were recruited from our existing clinical BCI program. Assent and consent were obtained in accordance with the Conjoint Health Research Ethics Board at the University of Calgary. Each family was provided with a kit of BCI equipment that included a basic laptop and a commercial-grade headset, the 14-channel *Emotiv EPOC X*. The EPOC X was used due to its ease of setup, with semi-dry electrodes and low-profile, flexible headband style design, as well as for its tolerability by participants. The pilot began in the summer of 2020 and has continued to expand since. The participating families ($n=7$ recruited over 18 months) had various prior experience with BCI, although none had previously used the equipment at home. Participating children were between the ages of 5 and 18 ($\text{avg}=11.6\pm2.3$ years), with QCP. BCI capacity was assessed by a pediatric neurologist (AK).

B. Iterative Software Design

Prior to the first iteration, families were surveyed to understand their context of home BCI use, including their capacity

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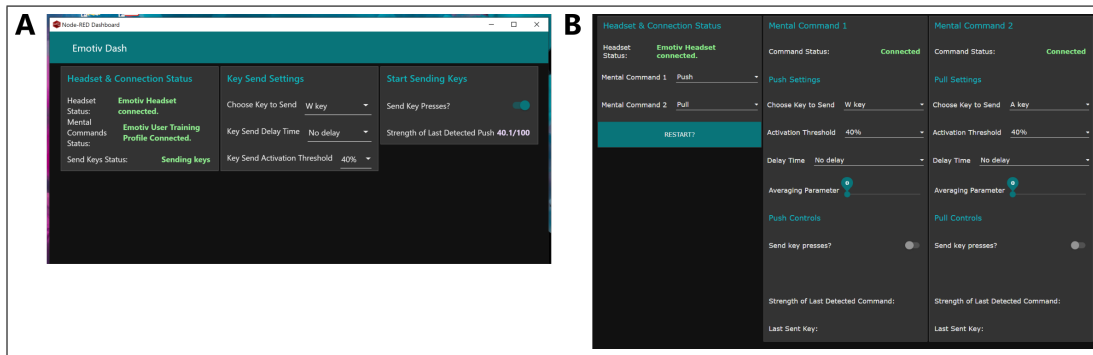


Fig. 1. User interface for the first two iterations of the home BCI software. Both iterations were created using Node-RED, a JavaScript-based visual programming tool. A) The first iteration, offering simple controls. B) The second iteration, offering multiple commands and more customizable settings.

for using complex technologies, the activities they wanted to use BCI with, and any adjustments needed in the home environment. Details of the home assessments and modifications (i.e., additional seating supports to help the child see the laptop screen) are beyond the scope of this paper. The first iteration of our home BCI software was prototyped based on user requirements established from these initial surveys. User requirements for subsequent iterations were identified based on family feedback. After deployment of each prototype, families were given the opportunity to set up and use the software independently as much as they wanted. The research team was available via virtual appointments to assist the families with any troubleshooting. Following an initial use period, families were asked for their feedback in the form of qualitative descriptions of likes/dislikes, challenges, and feature requests. Over the course of 18 months, 3 iterations of the software were designed, prototyped, and evaluated, each taking approximately 6 months to complete.

III. RESULTS

A. First Iteration: A Basic BCI Script

1) *Prototype Design:* The main user requirement for the first iteration was simplicity, in part so that it could be put together quickly so families could resume BCI use without significant delay after the onset of the pandemic, and in part to abstract away the complexity of BCI setup and use. Families most wanted to use BCI for their child to access entertainment in the form of computer games and painting. The resulting prototype was a simple script for integrating the BCI headset with keyboard-controlled applications, built using Node-RED, a JavaScript-based visual programming tool. The script translated “mental command” outputs (via imagined actions) from the headset’s proprietary API and mapped them to key bindings on the operating system. A simple UI let families adjust settings and suspend/resume key bindings. Basic visual feedback was also provided, demonstrating headset connection status (Fig 1).

2) *Family Experience:* Three families were enrolled in the home program to pilot this first prototype. All families were able to set up and use the BCI software independently. Troubleshooting and support requests at this stage were mostly related to setup of the hardware rather than the software.

As the families grew more proficient with the system, they requested greater personalization of parameters optimizing BCI control. Such parameters had previously been set by the research team, coded directly into the application.

3) *Implications For Next Iteration:* Node-RED offered a quick, straightforward way to quickly launch the home BCI program. It offered flexibility in the applications and services it could connect to. However, it was basic in the controls and customizability it provided, and families exceeded our expectations in their ability to master the simple controls in a short amount of time. Enabling customizability was the primary focus for the second software iteration.

B. Second Iteration: A Customizable BCI Interface

1) *Prototype Design:* Two major features were added in this iteration. First, families were given the ability to set their own mental commands (actions) to train for BCI use, allowing the children to explore and practice different commands independently. The second feature was enabling dual-command controls, mapping two commands to the keyboard simultaneously. Both added features reduced the reliance of the families on the research team, enabling them to make decisions about the child’s use of the system and minimizing the need for the researchers to adjust settings for them.

2) *Family Experience:* Two new families were enrolled in the program, resulting in a total of 5 families evaluating this prototype. Again, all families were able to set up and use the system independently. Families reported that the ability to explore and choose their own mental command increased their child’s feeling of control with the BCI. One technical challenge experienced at this stage related to the performance of the software when running concurrent applications (i.e., Zoom, games, etc.). Families also requested visual feedback to help their child monitor their effort when attempting imagery of their mental command.

3) *Implications For Next Iteration:* A significant drawback to the Node-RED interface emerged in this iteration; as more features were added there was a decrease in the performance in the software. Furthermore, maintenance and scaling through the visual programming interface became more complex with increasing time costs to add even small

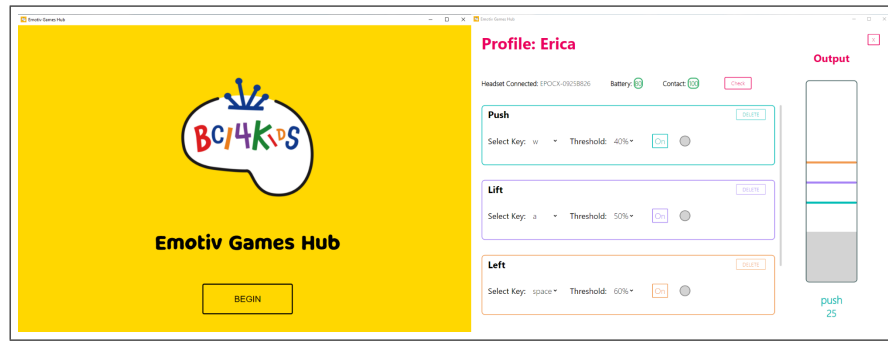


Fig. 2. The third iteration of the software, a custom desktop application built with Electron on the NodeJS runtime. The application allows further personalization and flexibility of BCI settings and provides the user with helpful visual feedback and clear labelling of controls.

features or changes. Therefore, for the next iteration, we moved to a custom-developed desktop application.

C. Third Iteration: A Standalone BCI Desktop Application

1) *Prototype Design:* For the final iteration we wrote a custom BCI software using Electron, a cross-platform desktop application package built on the NodeJS runtime. Electron allowed us to utilize the JavaScript, HTML and CSS programming languages for extensive flexibility in user interface design. The application, called the *Emotiv Games Hub*, was designed to be visually appealing and easy to navigate with intuitive controls (Fig 2). It built upon the previous feature requests from families to provide customizable control, including control of up to four different mental commands simultaneously and dynamically adjustable settings. The application also displayed connection status, battery power and contact quality (measured through impedance) of the headset's sensors. It was able to collect session metadata, user feedback, and brain activity data from the headsets. Performance issues from the previous iteration was resolved due to greater control over processing pipelines and software architecture. A custom application, although more complex to develop, offered greater opportunities for implementing user-centered features.

2) *Family Experience:* Three of the five families who had used the previous iteration(s) reported that they greatly enjoyed the look of the new interface and found it much easier to use. Additionally, two families expressed how helpful the visual feedback was for their child to practice their mental command. Two new families were also enrolled during this time and were able to use the software seamlessly. Most troubleshooting support during this phase involved headset fit, positioning, and signal quality. Support requests that did involve the software were related to upgrades in the headset API causing connectivity issues. Families requested personalization for the BCI calibration task, so they could employ strategies that could better motivate their child and help them focus on BCI control.

3) *Implications For Next Iteration:* We will continue to build on the *Games Hub* app for future iterations. We will next implement personalized calibration by providing customized visual feedback and auditory cues for practicing imagined actions. We will also integrate additional

commercial-grade headsets to resolve challenges concerning headset fit, positioning, signal quality and connectivity.

IV. DISCUSSION

We sought to design a software to support home use of BCI technologies for children with neurological disabilities. Our initial prototype focused on simplicity - families were provided with a commercial-grade headset employing a simple BCI paradigm, and the software prototype offered the most basic controls, abstracting away much of the complexity of BCI use. After the initial prototype was evaluated, it became clear that the families were capable of more nuanced control and customization. We adapted the software to provide additional settings, facilitating greater independence with the system. However, as we added more features, we reached the limits of the Node-RED interface in terms of efficiency and flexibility and moved to a custom-developed desktop application. Writing our own application introduced broad possibilities for user-centered features. This latest prototype is currently being used to facilitate independent home use of BCI systems for seven different families.

Kübler et al. [4] specified four stages of UCD for BCI development, which we applied to the design of our home software. First, we determined the families' *context of use* by surveying their technological capacity, determining activities they wanted to use BCI for, and assessing their home environment to identify required supports. Second, for each iteration we specified the *user requirements* by considering feedback from the previous prototype. Then, we produced *design solutions*, building the software prototypes, and finally *evaluated the designs* by giving the families the opportunity to use the software independently, receive direct support from the research team and report their feedback. In future work, we will formally assess the impact of UCD by evaluating standardized usability measures (i.e., effectiveness, efficiency and satisfaction [4]) and quality of life measures (i.e., Ped-sQL [14]) for our software.

The user-centered development of our home BCI software represents first steps in addressing the translational gap in BCI research for children. Specifically, we have focused on translating complex BCI "tasks" (i.e., system setup, calibration, and command generation) in a family-friendly way, integrated BCI control with desired end-applications, and



Fig. 3. Participants in the home BCI pilot program. Our home BCI software has provided new means for participation, creativity and self-expression for these children enabling them to create artwork, play games with their siblings, and express themselves on social media.

created a wide breadth of personalization to accommodate individual family needs. However, there is still much to be done before BCIs can be widely available for pediatric end-users. Most challenges faced by the families involved the hardware – issues with headset fit, signal quality, positioning, connectivity, durability, and integration with other medical equipment. A UCD approach to hardware design, like employed here with the software, would help mitigate these issues. Companies designing BCI headsets should involve end-users, including children and their families, in the design process early and often to create hardware that will maximize success in end-user environments.

A lack of knowledge about BCI also contributes to the translational gap – parents and clinicians typically do not know much about BCI [15], and this lack of knowledge can instill doubts in families about whether they are “capable” of using such a complex technology. More research needs to be done to understand how BCI skills, including system setup and operation, can be taught to both end-users as well as caregivers and anyone who will be assisting them with BCI use and interacting with them while the BCI is being used. Learning materials, format, and strategies will need to be personalized through a user-centered process as well.

V. CONCLUSION

BCI technologies, especially pediatric BCI technologies, face a critical translational gap, meaning few systems reach end-users in their daily lives. User-centered design can help address this gap by effectively matching the technology to end-user needs. We employed user-centered design principles to create a software that could support independent, long-term use of BCI technologies in the homes of seven families of children with quadriplegic cerebral palsy. Similar approaches are needed for all aspects of BCI technologies, including hardware, applications, and skill acquisition programs. User-centered design is essential to address translational challenges and realize the full potential of BCI systems, ultimately enabling new means of independence and participation for children with neurological disabilities.

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